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KONE: The MonoSpace® Launch in Germany

In November 1996 Raimo Hätälä, director of KONE Aufzug's new elevator business, was in the midst of planning the launch of his firm's latest product. The interim financial report he had just received in the mail confirmed that region-wide construction slumps and low differentiation among competitive offerings had led to significant price competition and margin erosion in the industry. KONE's operating income for the first eight months of 1996 was 6.0% of turnover, which compared with 6.7% for the same period in 1995. The report also projected that, absent significant changes, after-tax income for 1996 would be zero and worsen in the future.

To pull the firm out of the commodity rut Hätälä and other KONE managers were looking to the company's newest, revolutionary product, MonoSpace. Although news of the MonoSpace product had initially led him to exclaim to his colleagues, "With this, we can conquer the German market!," early test market and product launch results had given Hätälä cause for concern.

What, Hätälä wondered, was the size of the MonoSpace opportunity in Germany? How should he price and position MonoSpace? To what extent might MonoSpace cannibalize sales of KONE's existing low-rise elevators? What would be needed to ensure a successful launch? With more questions than answers before him, Hätälä began to review his options.

The Elevator Industry

Significant restructuring and consolidation in the late 1970s and 1980s found the worldwide elevator industry dominated in the early 1990s by five companies: Otis of the United States; Schindler of Switzerland; KONE of Finland; Mitsubishi Electric of Japan; and Thyssen of Germany (see **Exhibit 1** for more details on each competitor). Although they competed globally, these companies generally remained strongest in their domestic or regional markets. Toshiba and Hitachi of Japan and Goldstar of Korea were important competitors in the fast growing Asian market.

Numbers and types of elevators sold varied dramatically across the globe (see **Table A**), reflecting factors such as urbanization, population density, and government support for public housing.

Gordon Swartz of Market-Bridge and Professor Das Narayandas and prepared this case. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Table A Estimated Demand by Region for 1996 (units)

	Residential Low-rise	Other Low-rise	Mid-rise	High-rise	Total
Europe, Middle East, and Africa	65,000	8,500	4,000	500	78,000
North and South America	18,000	10,500	10,000	1,500	40,000
Asia and Australia	50,000	10,000	20,000	10,000	90,000
Total	133,000	29,000	34,000	12,000	208,000
Current Total Elevator Installed Base					>5,000,000

Source: Company records.

The elevator industry business was traditionally split into two sectors: new equipment and service that accounted for approximately \$9 billion and \$13 billion in global sales in 1995.

The traditional separation of product and service had generated interesting competitive dynamics in the elevator industry. Competition for new elevator installations was fierce leading to new elevator equipment often being sold at or below cost by the large competitors. Competition for elevator service contracts, on the other hand, was traditionally more orderly. Equipment suppliers usually had an advantage in winning contracts to service their installed bases. By tacit agreement, elevator companies maintained high margins on annual service contracts that were roughly equal to 5% of the purchase price of an elevator.

Low entry barriers due to the relatively simple electro-mechanical technology, steady demand and high margins in the service market had recently attracted many new competitors. These included small, local service-only providers that often enjoyed an advantage over the big manufacturers in terms of price, proximity and speed to service, important factors in the award of service contracts. Despite this trend, the large equipment suppliers had continued to do well given that approximately 80% of service contracts still flowed automatically from new equipment sales. However, there was some doubt that this situation would last for very long given the current economic environment.

Elevator Technology

Elevator technology varied dramatically with respect to travel height, traveling speed, ride comfort, machine room requirements, drive system, controls, cabin size, interior finishing, and price. Selecting an appropriate elevator technology often involved making multiple trade-offs that were all related to the type of drive system used to lift the elevator cabin.

Drive mechanisms The primary elevator drive technologies were: gearless (high speed) or geared (medium speed) traction (also called or “rope”); and hydraulic. Sales by type, subject to significant variations by region and country, were 10% for gearless, 30% for geared traction, and 60% for hydraulic. *Gearless traction* elevators, used primarily in commercial buildings, employed large, low-speed electric motors connected directly to drive pulleys to deliver the greatest ride comfort, travel height, and speed (2-12 meters per second). They were generally the only option for high-rise buildings (more than 25 floors). Being gearless, wear and tear and replacement costs were less than for geared traction elevators. *Geared traction* elevators, which employed a reduction gear between the motor and the drive pulley to move the cabin, provided moderate ride comfort, low to moderate travel height, and low to moderate speed (1.0-2.0 meters per second). Their speed was inadequate for high-rise buildings. Used only in low-rise buildings (less than 6 floors), *hydraulic* elevators offered minimal ride comfort, limited travel height, low speed (<0.6 meter per second), and could be priced

as much as 50% below substitutable geared traction elevators. Each hydraulic elevator needed 200+ - liters of oil that some elevator consultants considered to be a potential fire or environmental hazard.

Based on their performance characteristics and cost/benefit analysis, hydraulic elevators were suitable only for low-rise applications, gearless elevators for high-rise. Geared traction elevators had the widest application, primarily in mid- and low-rise, but occasionally in high-rise, buildings.

Machine room requirements The appendage-like shape of machine rooms, an inevitable component of elevator construction, made them difficult and costly to integrate into many buildings. They either occupied potentially useable building space in the basement or sat atop the shaft, rising above and marring the roofline. (**Exhibit 2** depicts the various machine room configurations.) In general, the taller the building, the larger the required machine room.

Gearless elevator machine rooms, always located on the roof above the shaft, ranged in size from 11 to 15 square meters per elevator. Geared traction elevator machine rooms averaged 11 square meters per elevator and offered three fixed placement options. The most common and least expensive, on top of the shaft (termed PT). The next most common location, on the lowest floor next to the shaft (termed PU), was generally more expensive because of more complex roping arrangements. The PU design was usually selected only if the machine room could not be placed on top of the building. The most expensive and least common selection, slightly above the top floor and to the side of the shaft (termed PS), involved the most elaborate roping arrangements. Hydraulic elevator machine rooms (termed PH), which averaged 5 square meters, could be placed on the lowest floor within 10 meters of the shaft.

Total elevator cost was roughly half equipment and half construction of the shaft and machine room and installation costs.¹ The geared traction elevator machine room typically represented approximately one-quarter of total elevator cost, hydraulic slightly less.

The Elevator Purchasing Decision

The complexity of elevator purchase decisions varied with building type and design. Generally, the taller, costlier and more complex a building, and thus the elevator system, the larger the number of people involved in the decision and factors to be considered. Selection of a high-rise commercial building elevator system, for example, might involve the property developer, building owner, construction contractor, architect, elevator consultant, and major tenants.

Prioritization of features and properties varied among individual participants, even within a class. Owners' decisions, for example, reflected their post-construction purpose. Owner/developers who intended to sell a building upon completion, tended to be most concerned about up-front costs. Owner/landlords were likely to care more about life-time costs, but, unless they could command a premium from their tenants, little about ride comfort and aesthetics. Owner/tenants, being involved throughout an elevator's life cycle, usually considered most factors.

A low-rise residential elevator purchase decision might involve from one to as many as five parties, the latter typically being property owner, construction company manager, architect, construction company purchasing agent, and building service manager.

¹ Elevators were typically installed by the general contractors that either used in-house talent or hired specialized sub-contractors such as elevator installers or elevator constructors to do the job.

KONE

KONE (pronounced *kô'-ne* and meaning “machine” in Finnish) was established in 1910. Originally focused on the repair and sale of rebuilt electrical motors, it expanded its business activities over the years to include the manufacture and sale of steel, maritime equipment, cranes, wood handling systems, and clinical chemistry analyzers. By 1995 KONE had divested its non-elevator businesses and become, through a series of 19 acquisitions, the world’s third largest elevator company, behind Otis and Schindler.

KONE’s elevator business was organized as two divisions: New Equipment, called V1; and Services, called V2. In 1995 KONE generated revenues of \$2.2 billion from sales of 16,500 new units and service contracts for 425,000 units. V1 accounted for 38%, V2 for 62%, of these revenues. Within V2, maintenance contracts accounted for 78% of revenues, modernization of existing elevators for 22%. (**Exhibit 3** presents KONE’s organizational chart. **Exhibit 4** summarizes the financials.)

KONE manufactured and sold a broad line of equipment, including standardized low-rise passenger elevators, medium-rise elevator systems, high-rise elevator systems, scenic elevators, hospital elevators, freight elevators, escalators and autowalks, and elevator components (see **Exhibit 5** for examples of KONE products.) Low-rise elevators accounted for approximately 75% of KONE’s equipment sales, mid-rise and high-rise elevators for 15% and 10%, respectively. In 1995 KONE spent approximately 1.5% of revenues on new product development.

With 90% of its sales outside Finland, KONE operated two headquarters, one in Helsinki and the other in Brussels. Sales by market in 1995 were: 53% EU; 4% rest of Europe; 29% North America; 10% Asia and Australia and 4% other countries.

KONE Aufzug

KONE Aufzug, which operated in Germany, continental Europe’s largest elevator market, generated revenues of approximately DM 206 million and profits of DM 13 million in 1995. The size of the German market and the volume of KONE’s sales of Europe made KONE Aufzug’s financial performance central to KONE’s overall success (See **Table B** for KONE Aufzug’s summary financials.)

KONE Aufzug was organized as a matrix of business divisions and geographical regions. The business divisions included V1 (new elevators), V2 (service), finance, and personnel; the three regions were North, South, and East. There were 25 local sales branches within the three regions. Each salesperson reported to a branch manager who, in turn, reported to both the regional director and the business division directors.

In 1996 KONE Aufzug employed 23 full-time and 20 half-time salespeople in V1 sales. Full-time salespeople averaged four to five half-day sales calls per week and spent the remainder of their time preparing proposals, answering queries from current and potential customers, and prospecting. The half-time salespeople divided their time equally between V1 sales and their other responsibilities; 13 were also branch managers and seven also worked as V2 salespeople. When describing their selling activities, branch managers were quick to remark that “Thyssen’s, Otis’, and Schindler’s sales forces each outnumber us by four or five to one.”

Forty-eight percent of 1995 sales were residential. Of these, 92% were PH; 6% PT; and 2% PU. Average prices for KONE’s standard 4-floor, low-rise, residential, volume-range elevator were: DM 60,000 for hydraulic PH, DM 75,000 for traction type PT; DM 80,000 for traction type PU; and

DM 120,000–DM 200,000 for traction type PS.² KONE's losses on new equipment sales averaged approximately 8% of sales for hydraulic, and roughly 5% of sales for traction elevators.

Table B KONE Aufzug Summary Financials (figures in DM 1,000s)

	1993	1994	1995
V1 Revenue	88,003	87,876	86,852
V1 Profit	-4,328 (4.9%)	-1,886 (2.2%)	-6,300 (7.3%)
V2 Revenue	114,718	116,762	118,628
V2 Profit	17,140 (14.9%)	18,140 (15.5%)	19,086 (16.1%)
Total Revenue	202,721	204,638	205,480
Total Profit	12,812	16,254	12,786

Source: Company records.

Evolution of KONE MonoSpace

A commercially viable, machine-room-less elevator had long been a compelling notion to elevator manufacturers, as it would yield significant additional usable space for revenue-generating purposes, and greater architectural freedom. In 1992 Otis Japan introduced a prototype machine-room-less elevator based on a linear induction motor, but its price premium exceeded its construction cost savings and revenue-generating possibilities and it was not a commercial success.

Building on the induction motor concept, a KONE R&D team in 1993 redesigned the motor geometry and used new materials to develop extremely thin, lightweight permanent magnets that eliminated the need for bulky, expensive components. Unlike comparable geared traction systems, that required a gearbox, KONE's new machine, the "EcoDisc," controlled speed by varying the frequency of alternating current supplied to the motor, as was commonly done in modern high speed gearless elevators. The EcoDisc thus offered ride comfort comparable to that of a gearless drive system. The need for a machine room was eliminated by placing the EcoDisc machine at the top of the shaft between one of the guide rails and the shaft wall; the entire elevator was thus said to occupy a "Mono" Space. The EcoDisc and MonoSpace are depicted in **Exhibit 6**.

The EcoDisc power unit also was extremely energy efficient, consuming only half the energy of comparable geared traction, and one-third of the energy required by comparable a hydraulic system. This and lower peak current translated into less expensive electrical wiring and fuses. Moreover, unlike hydraulic elevators the MonoSpace required no oil, eliminating potential fire and environmental hazards. Its installation time was approximately 190 hours, 60 hours less than for the simplest traditional elevator. (**Exhibit 7** compares the different drive systems.)

Theoretically, the EcoDisc was applicable to elevators across KONE's existing product line. But as elevator load and speed increased, so did the size of the machine, being eventually too large to fit within the shaft. With current EcoDisc technology a 16-person cabin required a machine room. Consequently, based on the most common cabin sizes KONE engineers had developed MonoSpace systems for 8-person and 13-person elevators with operating speeds of one meter per second that could be used in buildings of 12 floors or less. Concurrently, they were working to extend the capabilities of the EcoDisc machine and the MonoSpace product line.

² Average 1995 DM/\$ exchange rate was DM 1.43/\$1.00.

The MonoSpace Experience in Europe

KONE had targeted MonoSpace directly at Europe's largest new-equipment market segment, low-rise residential elevators. By the time Hätälä began developing a launch plan for Germany, MonoSpace had been test marketed in the Netherlands and officially launched, with varying degrees of initial success, in the Netherlands, France, and the United Kingdom. KONE managers selected the Netherlands because the market was dominated by low-rise elevators, KONE was the market leader, and the country's regulatory authorities, being relatively progressive, were likely to view the MonoSpace as a new elevator solution rather than one not in compliance with existing codes.³

Market Launches

Although construction was relatively stagnant in the three countries in which the MonoSpace was first officially launched, residential building accounted for roughly half of all construction activity (see **Table C** for more details on market size and KONE's market share in each country market). Approximately 90% of the units sold in France and 70% of those sold in the United Kingdom and the Netherlands were low-rise elevators.

Table C Unit Sales and Market Shares in 1995

	Units	KONE	Otis	Schindler	Thyssen	Others
France	7,000	14%	41%	20%	18%	7%
United Kingdom	3,300	20%	30%	10%	10%	30%
Netherlands	2,100	40%	19%	13%	6%	22%

Source: KONE's Director of Marketing Communications.

The U.K. market was unique in being dominated by one-star (low quality construction) and five-star (high-quality construction) buildings with little in between, driving demand for top-of-the-line and rock-bottom elevators. Many low-quality residential elevators were installed in the United Kingdom simply to meet regulatory requirements for access, particularly for elderly and handicapped persons. The preponderance of two- and three-star (medium-quality) buildings in France and the Netherlands drove broad demand for mid-range, mid-quality elevators.

Sixty to seventy percent of all elevators sold in the Netherlands were geared traction, a consequence of an anomalous market situation that had led to geared traction type elevators being about 5% less expensive than hydraulic elevators. With substitutable, low-rise geared traction elevators costing nearly twice as much as hydraulic elevators in the United Kingdom, hydraulic elevators accounted for 90%, geared traction elevators only 10%, of low-rise sales in that market. The French market was in-between, with 80% of low-rise elevators hydraulic and the remaining 20% geared traction.

³Because existing elevator codes had been written for installations with machine rooms, the MonoSpace, by definition, was not in compliance. Efforts were underway to promote EU standards, but the difficulty of modifying regulations varied dramatically from country to country. For example, because elevator regulations were part of Italy's constitution, to change them to allow a "machine-room-less" elevator would require an act of the Italian Parliament.

Pricing

Given KONE's differentiation and brand building objectives, the MonoSpace was generally priced in line with equivalent (and more expensive) geared traction elevators. Managers at KONE's Brussels headquarters suggested that the MonoSpace be priced above existing prices if KONE held less than 15% market share and in line with existing price levels if KONE otherwise (see **Table D**).

Table D Price Levels for KONE Low-Rise Elevators, 1996^a

	PH (Hydraulic)	PT (Traction)	PU (Traction)	MonoSpace
Netherlands	DG 65,000	DG 62,000	DG 68,000	DG 69,000
France	?	FF 150,000	?	FF 180,000
United Kingdom	£ 15,800	£ 30,000	?	£ 30,750

^aAverage 1995 currency exchange rates were: DG 1.60/\$1.00, FF 5.0/\$1.00, and £ 0.65/\$1.00.

The premiums exacted on the MonoSpace in the Netherlands and France were based on the KONE branch managers reasoning that a significant portion of the savings of *not* building a machine room would accrue to either the owner or construction company, motivating them to specify the MonoSpace. In France the MonoSpace's energy costs would be FF5,000 per year less than that of a comparable traction elevator, effectively repaying a FF30,000 premium in six years. The price in the United Kingdom was dictated primarily by the £15,000 transfer price to KONE U.K., which put it near the PT price.

Market Strategies

KONE viewed formal launches and articles in national and local specialist building and architectural journals as but a preamble to face-to-face, relationship-based selling, the activity it most relied upon to drive sales. To foster initial market acceptance, the MonoSpace was touted as a new drive system all other elements of which were identical to other KONE low-rise elevators. The style and scope of KONE's marketing activities varied according to the MonoSpace's projected sales success in the respective markets.

The Netherlands The MonoSpace was marketed in the Netherlands primarily through individual customer meetings. Approximately 3,500 architects, construction companies, owners, and consultants were invited via mailings to compare in one-on-one presentations a working MonoSpace with hydraulic and geared traction lifts. More than 100 such presentations were made within the first year of MonoSpace sales. MonoSpace-related articles published at the rate of approximately one per month subsequent to the launch each generated from 40 to 60 inquiries. Building specifications for the MonoSpace, supplied on disk to enable contractors and architects to "drop" this section into a building's plans, were due to be approved and included on the official Netherlands Building Design CD ROM, a resource used for most Dutch building designs.

France Letters announcing the MonoSpace to KONE's 22,000 existing French customers requested that they watch a television program that was to include a feature on the elevator. Concurrently, a MonoSpace advertisement was placed in a specialist building newspaper. The main market launch took the form of a series of breakfast meetings held in large cities across the country to which each office invited 20-30 guests, primarily architects, developers, building owners, owners or managers of smaller construction companies, and safety officials. Approximately 20 such breakfasts

were held during the first three months of sales. KONE salespeople also made individual on-site presentations at each of the country's six largest construction companies.

The United Kingdom Given the price sensitivity of the U.K. low-rise market, the MonoSpace was launched "to remind people that KONE is a technology leader." The underlying technology rather than low-rise elevator application was emphasized in a series of three presentations made at London's Science Museum. Five hundred construction companies, developers, quantity surveyors, consultants, and architects were invited; of 220 who accepted, 80 attended at least one presentation.

Sales Results

KONE's Netherlands managers projected that MonoSpace would account for 70% of total unit sales, and 100% of the units sold in the low-rise segment, within three years. MonoSpace reached the 70% target within the first 10 months of official sales and, within one year, KONE controlled 62% of the Netherlands low-rise market (up from 52% the year before) and just under 44% of the overall Netherlands market (up from 40%). KONE Netherlands was in the odd position of worrying about having gained too much market share and upsetting the market. "We didn't want to conquer the world with this," observed the Netherlands general manager,

we just wanted to maintain market share and get higher profits. We didn't want to scare the competition, because there is no way for us to eliminate an Otis or a Schindler. We'll have to live with these competitors forever. . . . Otis could buy KONE for cash . . . and can afford to drop its prices to the point that MonoSpace benefits—not having a machine room, energy savings, and so on—become meaningless.

Sales in France and the United Kingdom contrasted with the success in the Netherlands. In France, the first year sales target for MonoSpace was 300 units or 70% of KONE's annual residential sales. During the first three months, only 40 MonoSpace units were sold. In the United Kingdom no units had been sold one month after launch.

Customer reactions and learning points Notable among customers' generally extremely positive reactions to the MonoSpace were the oft heard "At last there is something new in the elevator industry" and "Why didn't you come up with this earlier?" Some aspects of the MonoSpace were more positively received than expected, others generated unanticipated worries. (**Exhibit 8** presents a summary of learning points.)

Competitor reactions Although minor spoiling tactics were encountered, KONE's director of V1 sales remarked that "the competition's reaction was one of stunned silence for the most part." One competitor offered to pay for the machine rooms if customers bought its elevators; another competitor told customers that there was no provision for ventilation with the MonoSpace, which was true, thermal losses being so low that there was no need for ventilation. In France some competitor salespeople told customers that KONE had received approval for only 10 MonoSpace elevator installations, which also was true; government authorities were to review the initial 10 installations and, if they proved acceptable, grant complete approval, standard procedure for the approval of any new technology in France.

Preparing for the German MonoSpace Launch

Pricing for the MonoSpace elevator in Germany had not been set, but production costs were estimated to be about the same as for a comparable hydraulic elevator.

The German Elevator Market

The German construction industry had undergone a cycle of boom and bust since the 1988 reunification of East and West. With the boom in construction the new elevator market grew from 8,000 units in 1988 to a high of 15,500 units in 1995, when the construction boom ended abruptly. Demand for new elevator equipment was now expected to shrink by 15% by the year 2000. The German elevator market was dominated by residential construction. The proportion of elevator units installed in low-rise residential buildings in 1995, 74%, was not expected to change significantly over the following five years. In 1995 hydraulic elevators accounted for approximately 60% of the German low-rise residential elevator market, geared traction elevators making up the rest. Two-thirds of the geared traction units were of the more expensive PU type. Demand for new commercial space, on the other hand, was dampened by significant over capacity.

Competition

The majors The six major players in the German elevator market were Schindler, Otis, Thyssen, KONE, Haushahn, and Schmitt & Sohn (see **Table E**). All operated throughout Germany and each maintained 24-hour service networks and new elevator sales and installation branches and manufacturing facilities both in Germany and abroad.

Table E German Elevator Industry: 1995 Market Shares

	New Elevator Market		Lifts in Service	Total Revenues
	Value	Units	Units	Value
Schindler	17.7%	19.4%	13.3%	21.1%
Otis	13.8%	11.6%	11.3%	19.4%
Thyssen	15.4%	12.9%	12.4%	18.1%
KONE	8.5%	9.2%	4.9%	6.7%
Haushahn	6.5%	5.8%	6.4%	5.6%
Schmitt & Sohn	5.4%	5.8%	3.3%	4.4%
Others	32.7%	35.5%	48.2%	25.0%

Source: Company records.

The mid-size players Approximately 30 mid-size players, with new equipment sales ranging from 100 to 300 elevators per year, operated regionally, although some of these produced a few key components (e.g., cars), most outsourced manufacturing.

The “cowboys” Small, local companies, termed “cowboys,” that usually operated within a single city numbered about 150. Most, lacking internal manufacturing capabilities, were focused on the purchase and assembly of components and installation and local service.

Market Performance

With the abrupt end of Germany’s construction boom, new elevator prices fell between 5% and 7% in 1994 and 1995. Many small and some mid-size players responded by abandoning or sharply curtailing efforts to sell new elevator equipment in favor of focusing on service, exerting additional price pressure there as well. Amongst the majors, Schindler’s reported losses were approximately

11%, Otis' 13%, of turnover. Further, Schindler had focused during this period on gaining share and become the clear market leader in hydraulic elevators. Otis' professed objective, to eliminate losses in the new elevator business, had caused it to lose market share.

Low-Rise Elevator Customers

Property developers, general contractors, and architects were among those involved in purchase decisions in Germany's low-rise elevator market. Property developers were principally concerned with the overall cost of developing a new building or renovating an existing property and factors that affected the investment value of their properties, including construction quality, timeliness of completion, and operating costs. Because choice of elevator was viewed as having little impact on overall construction costs, KONE and other elevator companies, perceiving property developers to rarely be involved in the decision, seldom communicated directly with them.

For buildings of all types, the general contractors responsible for construction and renovation according to property developers' and architects' plans exerted the greatest influence on elevator purchase. Although the four largest German contractors controlled approximately 20%, the construction market was highly fragmented, with nearly 20,000 small contractors vying for contracts.

Property developers consistently used the bid process, typically inviting bids from three or four, to pressure contractors for price reductions. In turn, contractors often used a competitive bidding process to procure specialized building systems such as steelwork, elevators, and HVAC (heating, ventilation, and air conditioning) systems. Occasionally, a two-stage bidding process was employed whereby contractors would invite "preliminary" bids, as from elevator suppliers, and incorporate a low-price offering into their overall construction bid, then, after the construction contract had been awarded, re-open the process and ask suppliers to resubmit bids. If the property developer and contractor had negotiated an overall price reduction, the contractor would try to pass on similar price reductions to the system suppliers.

Residential buildings were usually built by smaller contractors who, possessing little technical knowledge, often relied on architects to select elevators. Although architects generally did not make the elevator selection for the mid-size hotels and offices that larger contractors usually built, for which the higher end "residential" elevators were typically used, they almost universally selected elevators' cosmetic options (e.g., side- or middle-opening door, interior paneling material and colors, and so forth). KONE managers believed that in the German residential market the final elevator purchase decision was made by the general contractor 50% of the time, by the architect 40% of the time, and by a property developer 10% of the time.

KONE Aufzug's selling process

KONE Aufzug's selling process for new equipment had become well established over the years. In most cases (96% of purchases) customers initiated contact by sending elevator specifications and a request for bid to a local KONE branch. KONE being one of the major players, virtually all the customers included the firm in the shortlist of vendors to whom they sent out the request for bid. Thus, KONE had access to the entire demand for elevators in the German market.

Customer inquiries were followed by a visit from a KONE Aufzug salesperson to the customer contact, usually the construction company manager or architect. The salesperson reviewed the architect's drawings, design specifications, and any special requirements and would detail elevator options in the form of a sketch or CAD (Computer-aided Design) drawing.

In general, contractors wanted, at the lowest possible price, elevators that fit planned shaft dimensions and required no changes to architectural drawings. Purchase decisions were also influenced by the quality and relevance of the information the customer received, level of service, design of the bid document, and the customer's general impression of the salesperson. Observed one KONE branch manager: "The customer must feel that this salesperson alone is the expert in meeting the customer's needs." Once elevator specifications had been negotiated, a contractor's purchasing manager became the main point of contact and discussion shifted to payment terms and price. From start to finish, the procurement process ran 8-15 months.

Launch Decisions

Hätälä's pre-launch planning began with obtaining regulatory approval for the MonoSpace product. It also included establishing MonoSpace pilot installations in buildings across KONE Aufzug's three regions. In November 1995 four hydraulic elevator customers were given the option of converting their existing orders to MonoSpace. They were told that they would receive a new, improved drive unit at the previously agreed hydraulic elevator price, but, as the machine rooms for these buildings had already been designed and constructed, not that the MonoSpace technology eliminated the need for a machine room. Between January 1996 and June 1996 two salespeople were charged to arrange 30 more pilot installations. All pilot customers were given the option of switching from geared traction elevators to MonoSpace at no additional cost. The benefits of eliminating the machine room and reduced energy consumption, and the manner in which the roping worked, with an emphasis on feasibility and reliability, were carefully explained. All prospective pilot customers were, however, asked to "keep quiet about the technology."

Having consolidated his knowledge of the German market and lessons gained in the Netherlands, France, and the United Kingdom, Hätälä was concentrating on selecting the best marketing resources. He had at his disposal a marketing kit that had been developed by KONE's Brussels headquarters included the following components (**Exhibit 9** provides details of the marketing kit components and a summary of marketing resources).

Hätälä knew that German elevator companies, particularly KONE's larger competitors, relied on a broad range of marketing communications that included advertising, direct mail, customer and launch events, exhibitions, customer visits, and public relations. He wondered which of these were most appropriate for the launch of MonoSpace.

As he pondered these details, Hätälä recognized that immediate, favorable results from the German launch were vital to KONE. Yet, he had to keep in mind that the pricing and product positioning strategies he set for MonoSpace in Germany would have significant impact on the long-term prospects for KONE. With a lot at stake, he had little room for error.

Exhibit 1 Brief Descriptions of KONE's Major Global Competitors*Otis*

Founded in 1853, Otis, a wholly owned subsidiary of the United Technologies Corporation, was the global market share leader in the manufacture, sales, and service of elevators. In 1995 it sold more than 30,000 elevators and had 730,000 under maintenance contract. Otis employed 68,000 people worldwide in 17 production units and more than 600 sales offices in 45 countries. Revenues for 1995 were \$5.3 billion, up 14% from \$4.6 billion in 1994, and operating profits \$511 million, up 21% from \$421 million in 1994. Industry analysts attributed this increased profitability to aggressive and wide-reaching process re-engineering in the early 1990s (Otis had closed factories and reduced headcount). The company invested approximately 1.6% of annual revenues in R&D.

Otis was dominant in Europe, the United States, and Canada, with market share in these regions close to 30%. Its market share in Asia was approximately 20%. The company's aggressive, new-market entry strategy had made it the first foreign elevator company in the emerging markets of Asia and Eastern Europe.

Schindler

The Swiss engineering firm Schindler was ranked second in global elevator sales. Manufacture and sale of elevators and escalators accounted for 87% of the Schindler Group's 1995 revenues. Service accounted for 60% of these revenues, reflecting the company's shift during the 1990s from equipment sales to service. Although Schindler's total revenues had been flat in 1994 and 1995, approximately SFr 4.7 billion (\$4.0 billion) in both years, its 1995 after tax profits of SFr 78 million (\$67 million) were only one-half those of 1994. Most of Schindler's 20 production sites in 15 European countries had been converted from manufacturing to assembly. The company also operated more than 30 sales, maintenance, and installation facilities in 23 countries.

Schindler's expressed strategy was to expand its position in elevators and escalators and achieve an equal market split among its operations in Europe, America, and Asia/Pacific (the latter market requiring an aggressive growth plan). Schindler's profit strategy was to maintain its margins over volume and, to that end, avoid price wars.

Mitsubishi Electric

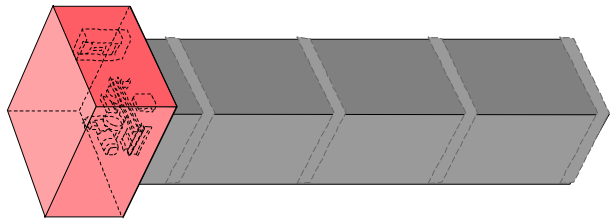
In 1995, 60 years after it began producing elevators, Mitsubishi Electric was Japan's leading elevator manufacturer, controlling more than 36% of the Japanese market. Turnover in 1995 was ¥2752 billion (\$27.8 billion), operating profit ¥177 billion (\$1.8 billion). Revenue broke down among the company's five divisions as follows: Consumer Products, 22%; Data Processing, 21%; Semiconductors, 20%; Industrial Equipment and Automation, 18%; and Heavy Electrical, including elevators, escalators, conveyors, and transformers, 24%. Particularly aggressive in Asia, Mitsubishi was the market leader in many Asian markets. In 1996 it opened a new Asian factory, doubled production in two existing Asian factories, and launched two joint ventures.

Thyssen

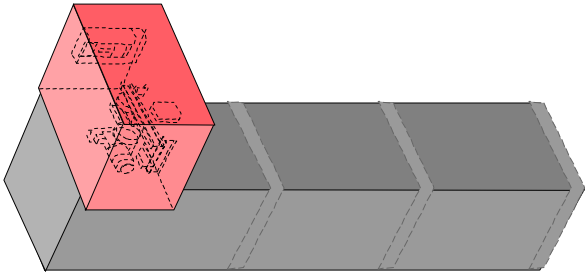
Thyssen Aufzüge, the world's fifth largest elevator manufacturer, was owned by Thyssen AG (1995 net sales of DM 10.1 billion (\$7.1 billion)). Thyssen Aufzüge's 1995 revenues were DM 2.2 billion (\$1.5 billion), up 5.2% from 1994. A decentralized operation, its subsidiaries exercised considerable autonomy over product ranges and sources. Thyssen Aufzüge manufactured only its most strategic components, outsourcing all others. Strong in Europe, with greater than 15% market share in 1995, but weak in North and South America, with a mere 2%, the company was investing heavily in Asia, having established manufacturing facilities in China and sales offices in China and Korea.

Source: KONE

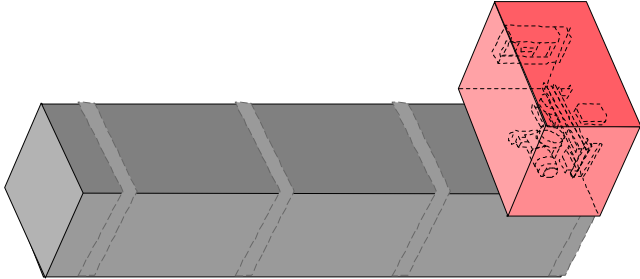
Exhibit 2 Elevator Machine Room Configurations



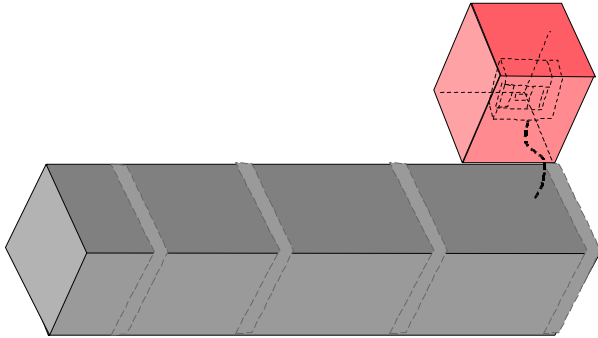
PT - Gearless or Traction



PS - Traction



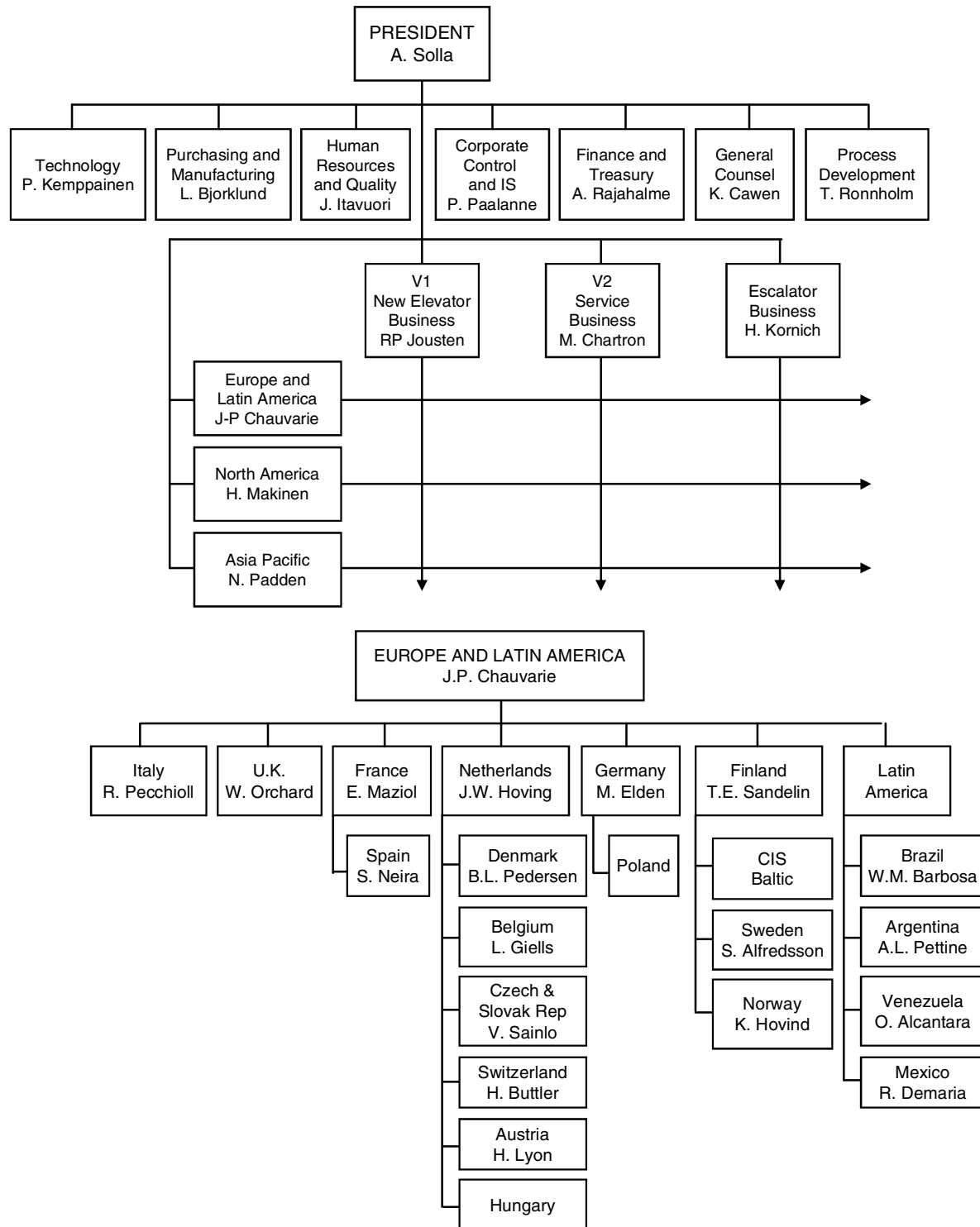
PU - Traction



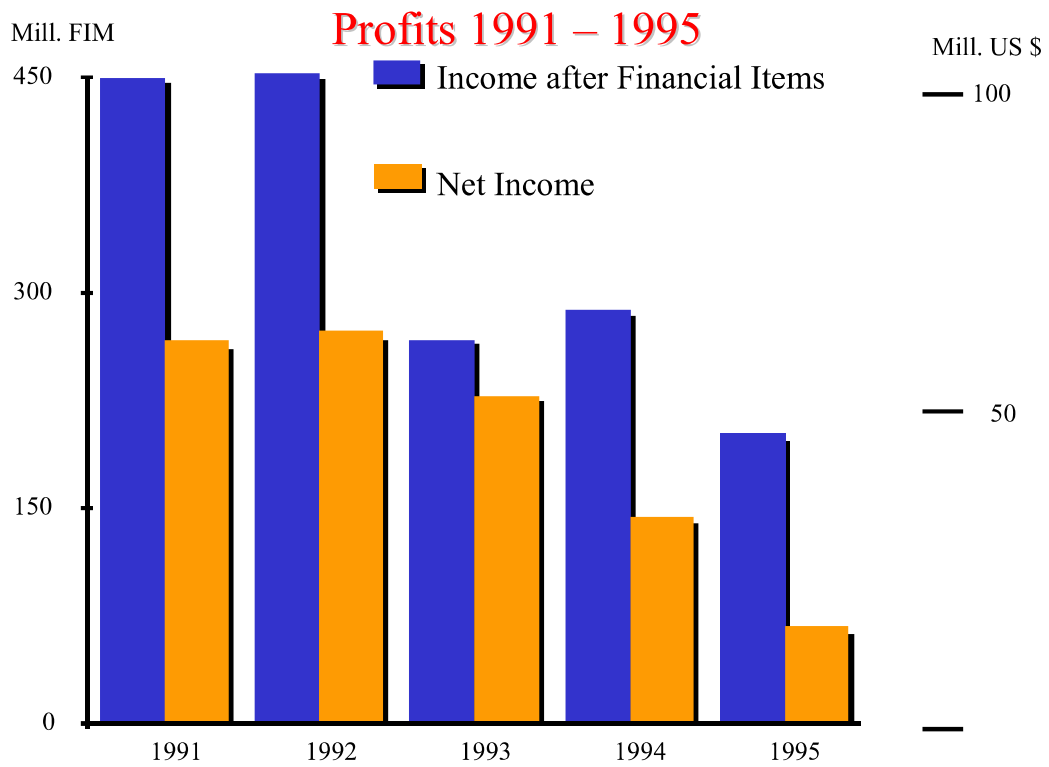
PH - Hydraulic

Source: KONE

Exhibit 3 KONE Organization Chart

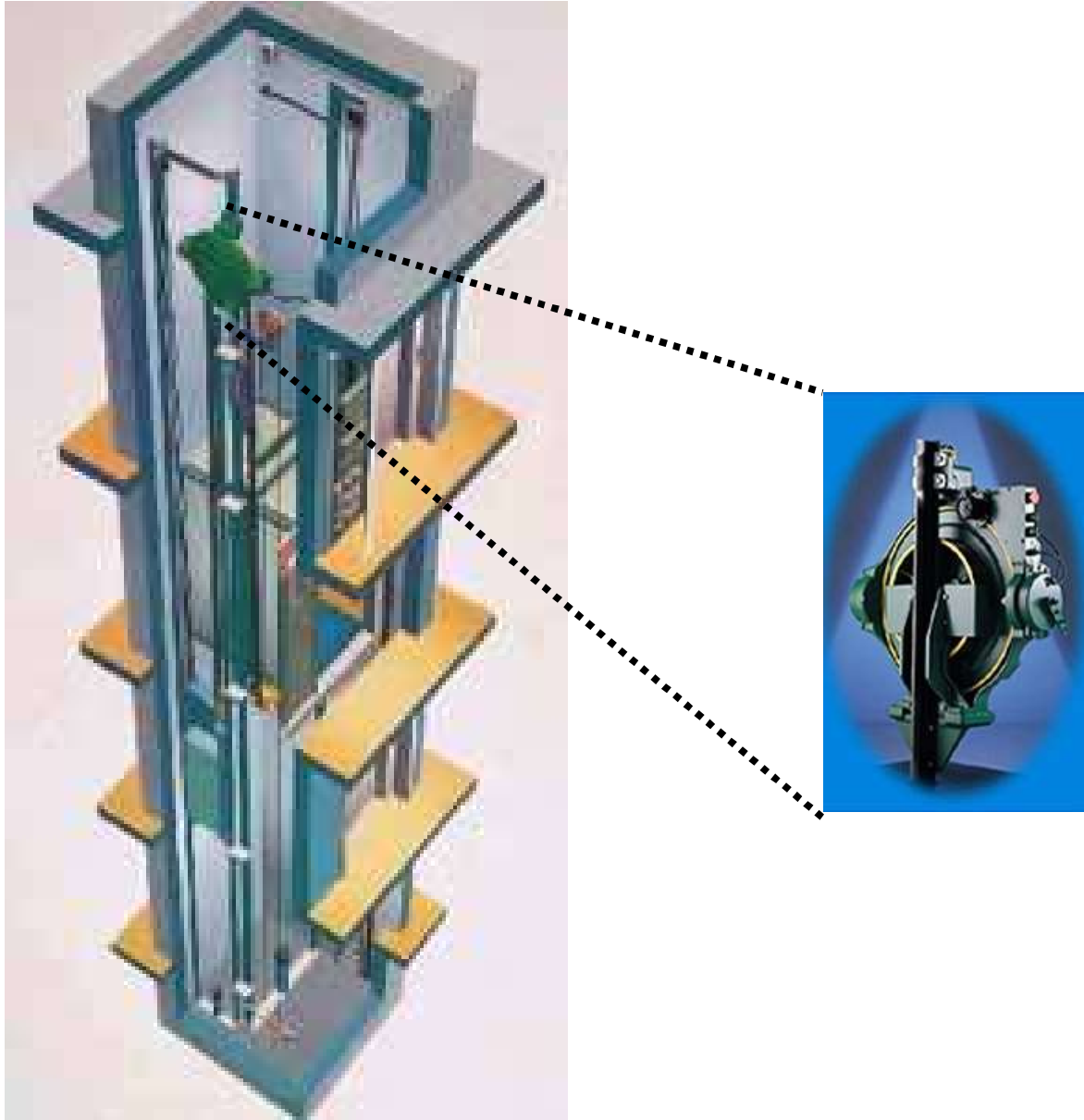


Source: KONE

Exhibit 4 KONE Five-year Financial Summary

Source: KONE

Exhibit 6 KONE's MonoSpace and EcoDisc



Source: KONE

Exhibit 7 A Comparison of Hydraulic, Traction, and EcoDisc Drive Units

Feature	Hydraulic	Traction	EcoDisc
Speed (mtrs/s)	0.63	1.0	1.0
Load (kg)	630	630	630
Motor size (kW)	11	5.5	3.5
Main fuse size (amp)	50	35	15
Energy consumption (kW _y)	7,500	5,000	2,500
Thermal loss (kW)	4.3	3.0	1.0
Oil requirements (Liters)	200	3.5	0
Weight (kg)	650	430	190
Machine room (m ²)	5	11	0

Source: Company records

Note: kW_y - kilo-watt years is a standard measure for comparing energy consumption of equipment.

Exhibit 8 Learning Points from European Marketing

Sales messages Construction companies, perhaps owing to their emphasis on initial price, proved the hardest sell. Most of the aspects expected to appeal to construction companies—elimination of the need for a machine room and for scaffolding or a crane and simplified installation processes—either were consequent to there being no machine room to be built, a savings that usually accrued to the owner, or were benefits not unique to MonoSpace, i.e., they were offered by the entire range of low-rise elevators.

Netherlands builders stood to save DG 7,000-DG 8,000 if a machine room already in the budget was not built. Otherwise the builder's savings would be nil. With the low-rise range builders rarely needed scaffolding and used a crane only if it was already on-site. Although less coordination needed between elevator installer and general builders, such process savings were generally not valued in the Netherlands. One-half day saved in the middle of a building process that took a year was considered inconsequential by construction companies that reasoned that their workers might spend that half day playing cards anyway. In France, however, where union regulations dictated a high degree of coordination among workers and weekly meetings of the *métier* heads, any process compression was valued. The situation was much the same in the United Kingdom.

KONE found that many small items, such as using lower risers, some time/process savings, etc., taken together, could amount to 5% savings of the construction and installation costs for the construction company.

Appeal of energy savings The MonoSpace's energy savings were not expected to be an important selling point in the Netherlands. Energy suppliers, in particular, however, found that the low-energy consuming MonoSpace didn't dim lights as geared traction or hydraulic elevators sometimes did as a consequence of the power surge required for take-off. The electrical fuses required by the MonoSpace were also much less costly: DG 60 per year for MonoSpace versus DG 1600 per year for hydraulic elevator fuses or DG 800 per year for geared traction elevator fuses. KONE Netherlands found that energy suppliers, an influence group it initially had not targeted, were consequently recommending MonoSpace.

Warning label Its drive unit and controller being located in the shaft on the top floor, the MonoSpace could not be used in buildings with penthouses (public access was necessary). Because its temperature had to be maintained at between 5°C and 40°C, it was also not suitable for outdoor use. KONE Netherlands discovered that architects had begun designing all their buildings, including those with penthouses and outdoor elevators, for the MonoSpace, that is, sans machine rooms. To avoid repercussions, all publicity for the MonoSpace carried a "warning" explaining these two limitations.

Construction company surcharge KONE discovered that in both France and the Netherlands some construction companies had retained the machine room savings by exacting a surcharge from the owner/developers. Consequently, all future literature directed to construction companies omitted the cost savings from eliminating the machine room and lower running costs of the MonoSpace.

Single supplier worries In all three countries, and particularly in the United Kingdom, customers worried that leaving the machine room out of building designs would leave them open to price gouging. Absent other machine-room-less suppliers, customers feared they had to pay whatever KONE asked. KONE countered by emphasizing its interest in long-term partnerships and preserving its reputation.

Example installations Customers wanted to see an installed MonoSpace before buying and few were willing to be "guinea pigs." This problem was addressed in the Netherlands by installing a working MonoSpace in KONE's Netherlands headquarters.

Source: KONE

Exhibit 9 Details on the Marketing Kit and Summary of Marketing Resources

The marketing kit developed by KONE's Brussels headquarters included the following components:

- A press kit featuring a CD-ROM and web site.
- A MonoSpace concept brochure to supplement existing low-rise line literature.
- Two PowerPoint sales presentations, one directed at builders, architects, and owners, the other a technical presentation for consultants and training purposes
- A 13-minute MonoSpace videotape.
- Trade press releases directed at architects, builders, and property owners/ managers.
- Building and planning guides that included dimensional sketches and a front-line CAD rendering of the MonoSpace.
- MonoSpace architectural specifications.
- A total elevator cost comparison form.
- A trade media advertisement on CD-ROM; a set of posters; a sample direct mail piece; a miniature static MonoSpace model.
- A sampling of promotional gifts such as pens and mouse pads.

The marketing resources available for the MonoSpace launch in Germany included the following:

Advertisements, which appeared mainly in elevator newspapers and journals (national) and architectural newspapers and journals (local and some national), were used primarily by small and mid-size companies. Large elevator companies seldom advertised. Single ads were most common, campaign rare.

Direct mail was targeted by large companies to architects, investors, and general contractors.

Launch events preceded new product introductions. All the large players had used "road shows" between 1993 and 1995. Customers were invited to local hotels for seminars and refreshments and received follow-up telephone calls.

Exhibitions were used mainly by component companies.

Customer visits supported by sales collateral were elevator companies' most common means of communication. Most literature had a technical slant, reflecting the average German customer's technical orientation.

Public relations activities were centered around press releases and press conferences. Otis and Schindler appeared regularly in the national and local press. Other companies generally received notice only in local papers or when something out of the ordinary occurred.

Exhibit 9. continued*Launch Options*

Direct mail—KONE's direct mail experience was limited. Ready-to-mail/fax response cards had achieved response rates of three to four per thousand. A mailing of 30,000 fax response cards, including list purchase and printing, cost DM 60,000.

Road show—The cost of 12 road shows—including two presentations per day, hotel space, catering, travel costs, equipment (portable model eco-disc and car), but excluding internal expenses (e.g., employees' time)—was roughly DM 350,000.

Sales visit—A sales visit usually required 1/2 day of a salesperson's time. Cost of materials and salesperson's time was roughly DM 500.

Video—Dubbing the video into German would cost DM 20,000, each video copy DM 5.

Telemarketing—One, two-to-three-minute telephone call to 30,000 people would cost approximately DM 100,000.

Seminar—For an audience of about 70, hotel space, catering, and the printing and mailing of invitations would cost approximately DM 10,000.

Exhibition/Trade show—To appear for three days at "The Konstruktor," a large, annual construction industry trade show held in Germany, would cost about DM 300,000. Previously, KONE had found that cost too high relative to the response it had elicited.

Trade press and journal advertising—Germany hosted approximately 25 architectural journals and about 12 building and construction journals.

- A black and white, one-page advertisement in a free monthly journal with a circulation of 25,000 would cost about DM 3,000.
- A black and white, one-page advertisement in a weekly journal priced DM 3.50 with a circulation of 17,000 would cost DM 2,300.
- A black and white, one-page advertisement in a weekly journal with a circulation of 31,000 cost DM 2,700.
- A black and white, one-page advertisement in a monthly journal with circulation 18,000 cost DM 1,600.

Source: KONE